(PV-SV-01) Software Validation Report

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Document Approval

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Revision History

Rev.	Date	Description
`0.0`	`2012.07.02`	Initial Version

Rev.	Date	Description
`0.1`	`2015.01.12`	User Interface Update
`0.2`	`2016.01.19`	New Function Implemented
`0.3`	`2017.01.13`	System Issue
`0.4`	`2018.01.12`	New Device Added
`0.5`	`2019.01.21`	Device Compatibility
`0.6`	`2020.01.30`	Device Compatibility
`0.7`	`2020.10.08`	GUI update
`0.8`	`2021.02.26`	Linkage usability improve
`0.9`	`2021.09.10`	Program Issue
`1.0`	`2021.10.08`	Device Compatibility
`1.1`	`2022.03.04`	Connection status improve
`1.2`	`2023.01.04`	Increasing image capacity
`1.3`	`2023.04.21`	Device Compatibility
`1.4`	`2023.08.29`	Update software version history
`1.5`	`2024.01.08`	Device Compatibility
`1.6`	`2024.02.23`	Document number changed from 603 to Z01 according to OP-709
`1.7`	`2025.08.14`	Revised according to the changed software architecture
`1.8`	`2025.09.09`	Update software version history
`1.9`	`2025.11.11`	Update software version history

1. Purpose

This document describes the software validation basis for Portview and demonstrates that the software supports its intended diagnostic workflow, safety objectives, and release expectations.

The report is intended to:

- describe the Portview software functions relevant to validation
- summarize the document set used to support validation
- identify the released software history and unresolved anomaly status
- provide the release conclusion for the covered software baseline

2. Scope

This report covers the Portview PC application used to acquire, view, manage, and export intraoral imaging data.

In scope:

- Portview feature description and software role
- lifecycle evidence relevant to validation
- verification and validation document references
- software version history
- unresolved anomaly status and release conclusion

The current released software history includes version `2.2.5.16`, dated `2024-10-02`, with Windows 11 and GenX-CR compatibility updates.

3. Referenced Documents

The following references support this validation report.

Reference	Use
Medical Device Regulation (EU) 2017/745	Regulatory framework
Quality management systems - Requirements	Quality-system reference
EN ISO 13485:2016 Medical devices Quality management systems	Governing quality management standard
EN 60601-1:2006/A1:2013 Medical electrical equipment	Basic safety and essential performance reference

Reference	Use
ISO 12052:2006 DICOM including workflow and data management	Governing interoperability standard
IEC 62304:2006/AMD1:2015 Software lifecycle processes	Governing software lifecycle standard
Guidance for the Content of Premarket Submissions for Software Contained in Medical Devices	Regulatory guidance
General Principles of Software Validation Final Guidance for Industry and FDA Staff	Regulatory guidance
Software and Medical Devices (VdTUV, 2001)	External guidance reference
QM Quality Manual	Internal quality reference
QM Design and Development Management Procedure	Internal process reference
QM Design Change Management Procedure	Internal process reference
QM Software Validation Procedure	Internal process reference
QM Service and Customer Complaint Procedure	Internal process reference
QM Document Management Procedure	Internal process reference
`PV-SV-02` Software Development Planning	Development planning reference
`PV-SV-03` Software High Level Design	Architecture reference
`PV-SV-04` Software Verification Plan	Verification planning reference
`PV-SV-05` Software Verification Report	Verification result reference
`PV-TM-01` Traceability Matrix	Requirement and risk traceability reference
`PV-CSRS-01` SwRS for Cybersecurity	Cybersecurity control baseline
`PV-NSE-01` Network Security Enclosure	Cybersecurity verification enclosure
`PV-FMEA-01` Risks FMEA	Authored residual-risk and hazard reference

4. Portview Software Description

Portview is an intraoral imaging application that acquires dental images, supports image viewing and patient-data workflows, and provides export functions for downstream use.

4.1 Device Feature

Portview acquires images from supported intraoral sensor systems and provides a user interface intended to support efficient 2D image review and diagnosis.

4.2 Role of Software

Portview acts as the parent software item for the user-facing PC application. Within the current scope, the principal Portview units are summarized below.

Unit	Description
Image Viewer	Displays images and provides zoom, measurement, annotation, and image-adjustment functions
Patient Management	Registers, updates, stores, loads, and manages patient information
Device Service	Manages modality connectivity and image acquisition workflows
Exportation Service	Exports images to supported formats such as DICOM files, plain image files, and removable media workflows

4.3 Roles And Responsibilities

Activity Area	Name	Position	Responsibility
Management	`K. Y. Ro`	CTO (Director) of R&D Center	Approves validation direction and released document set
Hazard Analysis	`C. H. Lee`	Deputy General Manager of R&D Center	Maintains hazard-analysis review
Hazard Analysis	`H. S. Park`	Deputy General Manager of R&D Center	Reviews risk and anomaly impact

Activity Area	Name	Position	Responsibility
Hazard Analysis	`S. R. Lim`	Manager of R&D Center	Supports risk-review and software assessment activities
Verification and validation testing	`J. B. Kim`	Staff of R&D Center	Executes and records validation-related activities
Documentation review	`C. H. Lee`	Deputy General Manager of R&D Center	Reviews controlled validation records
Documentation review	`H. S. Park`	Deputy General Manager of R&D Center	Reviews released validation content

5. Safety Classification And Level Of Concern

The software device is treated as a `Moderate` level of concern because a malfunction or design flaw may lead to an erroneous diagnosis or a delay in delivery of appropriate medical care that could lead to minor injury.

Within the Portview unit structure, the current design evidence supports Class `A` and Class `B` software units depending on the associated risk severity.

Classification Topic	Assessment
FDA level of concern	`Moderate`
Portview software-unit safety classes	`A` and `B`
Principal harm context	Delay of diagnosis and non-serious injury scenarios

For Class `B` software items, the design and validation approach is intended to ensure that the impact of software errors on users is minimized through architecture controls, rigorous verification, code review, and released validation evidence.

6. Lifecycle Evidence Set

The Portview validation basis is supported by the controlled document set below.

Area	Primary Evidence
Development planning	`PV-SV-02` Software Development Planning
High-level design	`PV-SV-03` Software High Level Design
Requirement definition	`PV-RS-01` RS for Portview, `PV-SRS-01` SwSRS for Portview
Design specification	`PV-SDS-01` SwSDS for Portview
Traceability	`PV-TM-01` Traceability Matrix
Verification planning	`PV-SV-04` Software Verification Plan
Verification execution	`PV-STP-01` SwSTP for Portview, `PV-STR-01` SwSTR-Z01 for Portview, `PV-TR-01` SwTR-Z01 for Portview, `PV-SYSTR-01` SystemTR-Z01
Cybersecurity controls	`PV-CSRS-01` SwRS for Cybersecurity, `PV-NSE-01` Network Security Enclosure
Risk management	`PV-FMEA-01` Risks FMEA and related risk-management references

6.1 Validation Evidence Structure

The validation basis is assembled from controlled lifecycle records rather than from a single test report alone. The evidence chain is expected to show that:

- development activities were planned and controlled
- requirements were documented and traced into design and verification records
- verification procedures were defined before execution
- released verification results support the final validation statement
- risk controls and residual-risk evaluation remain aligned with the covered release baseline

7. Validation And Verification Summary

Validation is supported by the lifecycle document set, requirement traceability, controlled verification procedures, released execution records, and residual-risk review.

Topic	Summary
Software development planning	Documented in `PV-SV-02`
Software requirements and architecture	Documented in `PV-SV-03` and controlled requirement specifications
Traceability	Maintained through `PV-TM-01` and linked validation records
Verification planning and execution	Covered by `PV-SV-04`, `PV-SV-05`, and released unit/integration/system records
Risk and anomaly review	Covered by `PV-FMEA-01`, cybersecurity references, and unresolved anomaly review

7.1 Validation Coverage

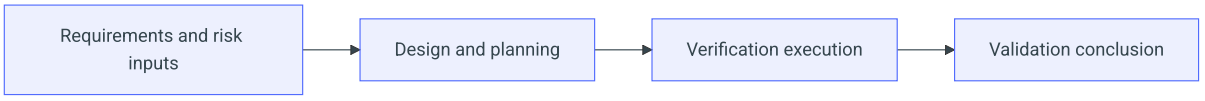
The validation basis covers the following lifecycle areas.

Validation Area	Coverage Statement
Software development planning	Controlled planning is documented in `PV-SV-02`
Software requirements	Requirement inputs are maintained through `PV-RS-01` and `PV-SRS-01`
Architecture and design	Architecture and decomposition are documented in `PV-SV-03` and `PV-SDS-01`
Verification planning	Verification planning and acceptance logic are documented in `PV-SV-04`
Verification results	Released execution outcomes are summarized in `PV-SV-05` and related procedure/result records
Risk and anomaly disposition	`PV-FMEA-01`, `PV-CSRS-01`, `PV-NSE-01`, and anomaly status are reviewed as part of validation closure

7.2 Risk And Residual-Risk Coverage

Coverage Topic	Controlled Reference
Product risk normalization	`PV-FMEA-01` Risks FMEA
Cybersecurity control baseline	`PV-CSRS-01` SwRS for Cybersecurity
Cybersecurity-focused verification	`PV-NSE-01` Network Security Enclosure
Requirement and result linkage	`PV-TM-01` Traceability Matrix

- `FR-xxx` risk items in `PV-FMEA-01` should remain traceable to requirements and released verification records.
- cybersecurity controls in `PV-CSRS-01` should remain linked to verification and residual-risk review.
- validation closure should not rely on unlinked or title-only risk references.



8. Software Version History

The released Portview software history recorded for validation is summarized below.

Version	Date	Reason Of Change	Remark
`1.0.0.0`	`2012.08.13`	Firstly prepared	
`2.2.5.4`	`2020.02.24`	Reduce sensor noise	
`2.2.5.5`	`2020.03.09`	Improve sensor recognizing	
`2.2.5.6`	`2020.05.11`	Improve image processing	
`2.2.5.7`	`2020.08.19`	Change images sequence	
`2.2.5.8`	`2020.10.06`	Add full series mount	

Version	Date	Reason Of Change	Remark
`2.2.5.9`	`2021.03.02`	Resolve the mount issue	
`2.2.5.10`	`2021.09.10`	Add software filter	
`2.2.5.11`	`2021.10.25`	Display UDI code	
`2.2.5.12`	`2022.03.04`	Connection sequence update	For GTIS sensor
`2.2.5.13`	`2022.12.28`	Add FMX32 mount	
`2.2.5.14`	`2023.03.02`	Data folder information	
`2.2.5.15`	`2024.01.10`	Improve modalities' connectivity	
`2.2.5.16`	`2024.10.02`	Add newer OS (Windows 11), GenX-CR compatibility	For GenX-CR
`2.2.5.17`	`2024.11.26`	Fix bugs, refactoring internal codes	Internal version
`2.2.5.18`	`2025.02.19`	Add sensor support	For Handy sensor; internal version
`2.2.5.19`	`2025.09.09`	Fix bugs, refactoring internal codes	
`2.2.5.20`	`2025.11.11`	Apply requests from GAI department	

9. Unresolved Anomalies

No Portview-specific unresolved anomaly table is currently embedded in this document body. Any unresolved anomaly carried into release should be linked to the controlled anomaly list, risk evaluation, and user communication records.

Item	Status	Notes
Portview unresolved anomaly summary	`No anomaly table embedded`	Link controlled anomaly records if applicable
User communication requirement	`Applicable when unresolved anomalies remain`	Device labeling and user communication should reflect approved dispositions

9.1 Anomaly Governance

The validation conclusion assumes that:

- unresolved anomalies affecting safety or effectiveness are identified and evaluated before release
- approved dispositions are reflected in the controlled anomaly and risk records
- user communication is updated when unresolved anomalies require operational awareness after release

10. Conclusion

The Portview software has been evaluated through the planned verification and validation activities, the identified risks have been reduced to acceptable levels through the applicable controls, and the residual-risk balance supports release of the covered software baseline.

The validation basis supports release on the following grounds:

- the controlled lifecycle document set establishes planning, design, verification, and release evidence
- the software history and covered release baseline are identified within the controlled records
- the available anomaly and risk information does not identify a Portview-specific unresolved issue that blocks release

11. IEC 62304 Compliance Summary

The Portview document set is intended to support compliance with the applicable IEC 62304 lifecycle clauses through the controlled records listed below.

IEC 62304 Topic	Primary Deliverable Mapping
Software development planning	`PV-SV-02` Software Development Planning
Software safety classification	`PV-SV-01` Software Validation Report

IEC 62304 Topic	Primary Deliverable Mapping
Software requirements analysis	`PV-RS-01`, `PV-SRS-01`, and `PV-TM-01` traceability records
Software architectural design	`PV-SV-03` Software High Level Design
Software detailed design and implementation	`PV-SDS-01` and controlled build and implementation records
Software verification	`PV-SV-04`, `PV-SV-05`, and released procedure/result records
Software risk management	`PV-FMEA-01`, `PV-CSRS-01`, `PV-NSE-01`, and related risk-management files
Software configuration management	`PV-SV-02` planning records and controlled configuration items
Software problem resolution	Design change and maintenance/problem-resolution records

Open Items

- `Conditional:` Link the controlled unresolved anomaly list if unresolved release items exist for the covered baseline.
- `Conditional:` Link the traceability matrix and released verification records directly if the released document set requires explicit in-document references.